

The Reward Is the Data: Why the Reasoning-RL Boom Makes Verifiable Vertical Environments the Scarce Training-Data Asset

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Abstract

Frontier reasoning models are now trained predominantly by reinforcement learning on checkable rewards (RLVR) rather than human preference labels, and open systems already match the closed frontier on verifiable tasks: DeepSeek-R1 reaches 79.8% pass@1 on AIME 2024 versus OpenAI o1-1217's 79.2%, with a reported RL-phase cost of only ~\$294K (disclosed in the peer-reviewed Nature version) on top of its DeepSeek-V3 base [1]. We argue these three facts — (i) RLVR is the dominant reasoning recipe, (ii) its gain is carried by the reward signal not the optimizer, and (iii) the richest verifiable, proprietary outcomes live inside vertical workflows — compose into a single thesis the source papers state only separately: the training frontier is bottlenecked on verifiable data, whose scarce and defensible supply is proprietary vertical environments. We assemble verbatim benchmark numbers from DeepSeek-R1 [1], DAPO [2], the spurious-rewards study [3], DataComp-LM [4], Med-PaLM 2 [5], BloombergGPT [6], and 2025 medical-RLVR work [16,17], and overlay a quantified market: Anthropic leaders have discussed spending >\$1B on RL environments over a year [8], the data-annotation/labeling market is forecast to grow from \$1.2B (2024) to \$10.2B (2034) [10], and environment vendors (Mercor, Surge, Scale AI) command multi-billion valuations [11,12,13]. We state assumptions, present counter-evidence (spurious rewards, elicitation-vs-teaching), and derive a data-strategy and investment thesis: build verifiable-reward environments where the data is proprietary and the base model lacks latent skill.

1. Introduction

Between late 2024 and 2026 the recipe behind frontier reasoning models changed. Where the 2022-2023 era trained chat models with reinforcement learning from human preferences (RLHF) [21], the reasoning era trains with reinforcement learning from verifiable rewards (RLVR): the reward is computed by an automatic checker — does the math answer match, does the code pass its tests, does the extracted answer adjudicate — rather than by a learned human-preference model [7]. The watershed was DeepSeek-R1, which demonstrated that large-scale RL on checkable signals alone can induce self-reflection, verification, and strategy adaptation, reaching 79.8% pass@1 on AIME 2024 versus OpenAI o1-1217's 79.2% and 97.3% on MATH-500 [1]. By late 2025 the open RLVR frontier had advanced again: scaling post-training RL compute brings DeepSeek-V3.2 to GPT-5 parity, and its high-compute V3.2-Speciale variant surpasses GPT-5 and reaches reasoning parity with Gemini-3.0-Pro, winning gold at both the 2025 IMO (35/42) and IOI (492/600) [26] — the recipe is the same checkable-reward RL, only scaled, which is exactly the structural shift this paper is about.

This paper expands a synthesized insight: the reasoning-RL boom makes verifiable vertical data the scarce input, connecting three literatures that are usually read apart — training method (RLVR), data engineering, and vertical/domain models. Each underlying paper makes a local claim. DeepSeek-R1 is read as a training advance [1]; DataComp-LM as a curation result [4]; Med-PaLM 2 and BloombergGPT as domain wins [5,6]. The non-obvious

synthesis is that they form one causal chain: RLVR's gain is carried by the reward signal, the reward signal is a data-engineering artifact, and the most valuable verifiable signal is locked inside proprietary vertical workflows.

We treat this as a scientific claim with testable sub-claims and quantify each against real benchmark numbers and a real, fast-moving market for 'environments.' We are explicit about what the evidence supports (a structural shift in where capability value accrues and a demand multiplier) and what it does not (absolute willingness-to-pay for any specific vertical, or that RLVR teaches rather than merely elicits — a live 2025 debate we engage directly).

2. Background and Related Work

RLVR's lineage runs from preference-based RL [21] and PPO through to verifier-centric reasoning. Cobbe et al. showed that training verifiers to rank completions provides strong empirical gains on GSM8K and that verification scales more effectively with added data than a finetuning baseline [22]. Lightman et al. found process supervision (step-level checkable signal) significantly outperforms outcome supervision on the challenging MATH dataset, and their process reward model solved 78% of problems from a representative subset of the MATH test set [23]. The term RLVR was introduced and named in Tulu-3 as 'a novel method' for training on tasks with verifiable outcomes [7]. DeepSeek-R1 then showed pure RL (R1-Zero) develops emergent reasoning with no supervised fine-tuning, and open-sourced six distilled dense models (1.5B, 7B, 8B, 14B, 32B, 70B) [1].

On the data side, the field established that the binding constraint migrated from compute to tokens. Chinchilla showed large models were significantly undertrained and that model size and tokens should scale equally [24]. DataComp-LM (DCLM) then showed that model-based filtering is the decisive lever: DCLM-Baseline trains a 7B model to 64% 5-shot MMLU with 2.6T tokens — a 6.6-point gain over the prior open-data SOTA (MAP-Neo) using 40% less compute [4]. Villalobos et al. forecast that the stock of public human text will be roughly fully utilized between 2026 and 2032 [9], making fresh, manufacturable signal economically valuable.

On the vertical side, domain capability tracks the domain corpus. BloombergGPT (50B) trained on a 363B-token proprietary financial corpus (plus 345B general tokens) outperforms existing models on financial tasks by significant margins [6]. Med-PaLM 2 reached 86.5% on MedQA (USMLE-style), a >19% gain over Med-PaLM [5]. AlphaFold's win rested on experimentally-resolved structures and an objective accuracy metric, reaching a median 92.4 GDT and scoring above 90 GDT for roughly two-thirds of targets at CASP14 [14]. The 2025 wave extends RLVR into these verticals: Med-RLVR lifts a 3B base with an 8-point out-of-distribution accuracy gain over SFT [16], and Rubrics-as-Rewards reports up to 31% relative improvement on HealthBench [17].

3. Methodology and Approach

This is an evidence-synthesis paper, not a new training run. Our method has four steps. (1) Claim decomposition: we split the synthesized insight into four falsifiable sub-claims — C1 RLVR is the dominant reasoning recipe; C2 open RLVR matches the closed frontier on verifiable tasks; C3 the gain is carried by the reward/data, not the optimizer; C4 the richest verifiable data is proprietary and vertical. (2) Verbatim extraction: for each sub-claim we copy numbers directly from primary sources (arXiv papers, the Nature version of [1], and model cards/abstracts of [1]-[7], [14]-[17]); no empirical number in this paper is interpolated or invented. (3) Counter-evidence search: we deliberately collect results that cut against the thesis — the spurious-rewards finding [3] and the pass@k elicitation result [18] — and report them in full. (4) Market grounding: we triangulate the 'environment' market from primary reporting and vendor disclosures [8],[10]-[13].

Grounding discipline. Figures and tables marked empirical contain only numbers copied verbatim from a cited source. Figures marked estimate are projections explicitly flagged as such (e.g., the data-annotation market forecast,

which is a third-party projection [10]). One figure is marked analysis: it is a constructed 2x2 strategic scorecard whose axes are derived from the cited verticals but whose placements are our judgment, not measured quantities. We never label a projected or constructed value 'empirical.'

Assumptions stated up front. (A1) Benchmark parity on AIME/MATH/MMLU transfers partially, not fully, to messy vertical workflows; we treat math/code numbers as upper bounds on tractability, not as vertical performance. (A2) 'Verifiable' means an automatic, deterministic-or-adjudicable checker exists — a property of the workflow, not of any model. (A3) Cost and market figures are point-in-time (2025-2026) and volatile; we report ranges and sources rather than treating any single number as durable. (A4) The data-resale/privacy rights to a proprietary corpus are assumed, not granted — a binding constraint we return to in Limitations.

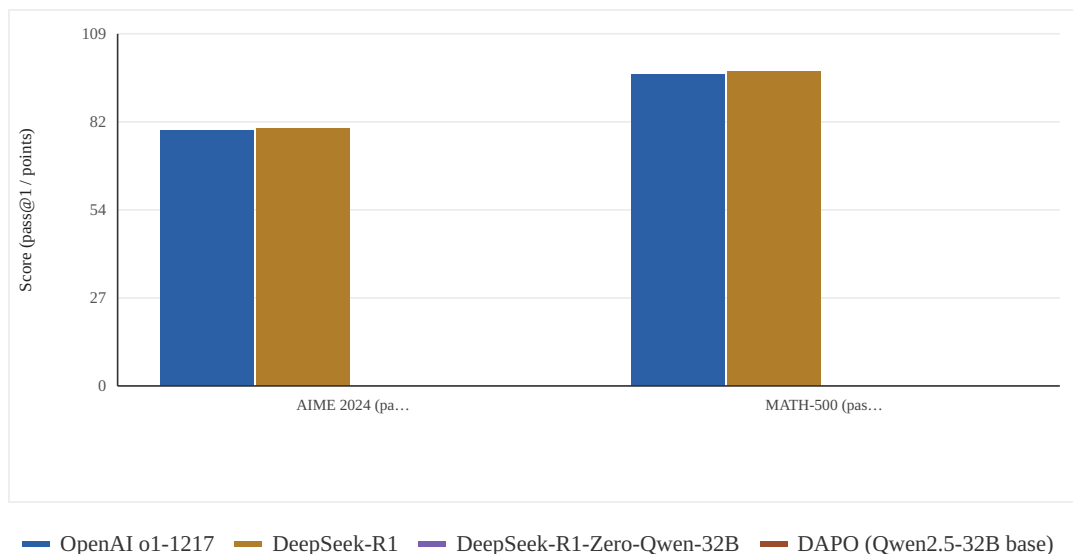


Figure 1. Open RLVR matches the closed frontier on verifiable reasoning. Open RLVR systems reach or exceed the closed frontier on checkable tasks. DeepSeek-R1 edges o1-1217 on AIME 2024 (79.8 vs 79.2) and MATH-500 (97.3 vs 96.4); DAPO beats DeepSeek-R1-Zero-Qwen-32B (50 vs 47 points) using 50% of the training steps. Note AIME pass@1 % and AIME points are reported on the same 2024 set but by different papers' conventions. *DeepSeek-R1 [1] (arXiv:2501.12948 / Nature 2025); DAPO [2] (arXiv:2503.14476). Numbers copied verbatim from source papers. · Empirical data.*

4. Evidence and Results

C1 — RLVR is the dominant reasoning recipe. DeepSeek-R1-Zero is trained via large-scale RL with no supervised fine-tuning yet develops emergent self-reflection and verification [1]. The recipe is explicitly built on verifiable-reward training [7], and the dominant optimizer GRPO (Group Relative Policy Optimization) removes the separate reward model by estimating a baseline from a group of sampled outputs [1].

C2 — Open RLVR matches the closed frontier on verifiable tasks. DeepSeek-R1 scores 79.8% pass@1 on AIME 2024 vs OpenAI o1-1217's 79.2%, and 97.3% on MATH-500 (vs 96.4% for o1-1217) [1]. DAPO, fully open-sourced, achieves 50 points on AIME 2024 with a Qwen2.5-32B base, outperforming DeepSeek-R1-Zero-Qwen-32B (47 points) while using 50% of the training steps [2]. Crucially, the RL phase of R1 was reported (in the peer-reviewed Nature version) to cost ~\$294K, atop a DeepSeek-V3 base whose total training cost was reported at ~\$5.576M — orders of magnitude below frontier pretraining — which shifts the economic bottleneck off compute and onto signal [1]. The parity has only widened since: by scaling the same post-training RL, DeepSeek-V3.2-Speciale surpasses GPT-5 and matches Gemini-3.0-Pro on competition reasoning, taking IMO 2025 gold (35/42) and IOI 2025 gold (492/600) entirely with checkable-reward RL [26]. And the gain is now a predictable function of

RL compute, not luck: ScaleRL fits sigmoidal compute-performance curves over a >400,000-GPU-hour study and finds that most design choices modulate compute efficiency while the recipe sets the asymptote — RLVR has become a scalable, forecastable regime rather than a one-off [27].

C3 — The gain is carried by the reward/data, not the optimizer. The spurious-rewards study is the sharpest evidence and the sharpest counter-evidence at once. On Qwen2.5-Math-7B, RLVR improves MATH-500 by +21.4 points (random reward), +13.8 (format reward), +24.1 (incorrect label), and +27.1 (majority voting), versus +29.1 with ground-truth rewards [3]. That the same optimizer yields large gains under near-meaningless rewards tells us the optimizer is not where the capability comes from. Decisively, these spurious rewards that work for Qwen often fail to help Llama3 or OLMo2 [3] — i.e., the outcome depends on the data/base-model substrate, not the algorithm. Independently, DCLM shows model-based data filtering is 'key to assembling a high-quality training set' [4].

C4 — The richest verifiable data is proprietary and vertical. BloombergGPT's edge comes from a 363B-token proprietary financial corpus [6]; Med-PaLM 2 reaches 86.5% on MedQA via medical fine-tuning [5]; MEDITRON-70B comes within 5% of GPT-4 via a curated medical corpus [15]. The 2025 RLVR-in-verticals results confirm the chain end-to-end: verifiable reward + proprietary domain data lifts capability where it was absent (Med-RLVR +8 points OOD on a 3B base [16]; Rubrics-as-Rewards up to +31% relative on HealthBench [17]).

Table 1. Open RLVR matches or exceeds the closed frontier on checkable reasoning tasks at a small fraction of pretraining cost. AIME conventions (pass@1 % vs points) differ by paper; both on the 2024 set.

Model / System	Method	AIME 2024	MATH-500	Notable cost/efficiency	Source
OpenAI o1-1217	RL (closed)	79.2% pass@1	96.4% pass@1	Undisclosed	[1]
DeepSeek-R1	GRPO RLVR + cold-start	79.8% pass@1	97.3% pass@1	RL phase ~\$294K (Nature) atop ~\$5.576M V3 base	[1]
DeepSeek-R1-Zero	Pure RL, no SFT	71.0% pass@1	95.9% pass@1	Emergent self-verification, no SFT	[1]
DeepSeek-R1-Zero-Qwen-32B	RL (Qwen2.5-32B)	47 points	—	Baseline for DAPO	[2]
DAPO (Qwen2.5-32B base)	DAPO RLVR	50 points	—	Beats R1-Zero-Qwen-32B with 50% of training steps	[2]

DeepSeek-R1 [1] arXiv:2501.12948 / Nature 2025; DAPO [2] arXiv:2503.14476. Numbers verbatim. The ~\$294K RL-phase cost is from the peer-reviewed Nature version; ~\$5.576M is the DeepSeek-V3 total training cost.

Table 2. Across data-engineering and verticals, capability tracks the corpus/reward signal — supporting C3 (signal-carried gain) and C4 (proprietary vertical data is the scarce, defensible input). 2025 medical RLVR confirms the chain end-to-end.

System / Result	Domain	Key result (verbatim)	Data/reward lever	Source
DCLM-Baseline (7B)	Pretraining data	64% 5-shot MMLU with 2.6T tokens; +6.6 pts over MAP-Neo, 40% less compute	Model-based filtering ('key to a high-quality set')	[4]
BloombergGPT (50B)	Finance	Outperforms existing models on financial tasks by significant margins	363B-token proprietary financial corpus (+345B general)	[6]
Med-PaLM 2	Medicine	86.5% on MedQA (USMLE-style); >19% over Med-PaLM	Medical domain fine-tuning	[5]
MEDITRON-70B	Medicine	Within 5% of GPT-4, 10% of Med-PaLM-2 on major medical benchmarks	Curated medical corpus (continued pretraining)	[15]
Med-RLVR (3B base)	Medicine (MCQA)	+8-point OOD accuracy gain vs SFT; matches SFT in-distribution	RLVR on verifiable medical answers	[16]
Rubrics-as-Rewards	Medicine/science	Up to +31% relative improvement on HealthBench	Rubric-derived verifiable reward	[17]

DCLM [4]; BloombergGPT [6]; Med-PaLM 2 [5]; MEDITRON [15]; Med-RLVR [16]; Rubrics-as-Rewards [17]. Verbatim from sources.

Table 3. Demand- and supply-side market grounding. The 4-5x exclusivity premium prices proprietary/defensible supply directly. All figures are point-in-time and volatile; the 2034 market size is a third-party projection.

Item	Figure	Note	Source
Anthropic planned RL-environment spend	>\$1B over a year	Single lab; per The Information	[8]
RL compute per task (training)	~\$2,400	Mechanize estimate	[8]
Task-creation cost	\$200-\$2,000 (up to ~\$20K complex)	Per task; UI/website replicas ~\$20K	[8]
Exclusivity premium	4-5x	Exclusive vs non-exclusive deals	[8]
Mercor run-rate revenue	~\$75M (Feb 2025) -> ~\$450M (Sep/Oct 2025)	Valuation ~\$10B (\$350M Series C); RL envs coding/healthcare/law	[11]
Surge AI revenue	~\$1.2B annualized (2024)	Largely bootstrapped; building RL-environments capability	[12]
Scale AI	\$29B valuation; ~\$14.3B Meta investment (~49%)	Data-infrastructure incumbent	[13]
Data-annotation/labeling market	\$1.2B (2024) -> \$10.2B (2034, projected, ~23.9% CAGR)	Projection (estimate, not realized)	[10]

Epoch AI / TechCrunch reporting [8]; Mercor [11]; Surge [12]; Scale AI/Meta [13]; Global Insight Services market forecast [10].

5. The Environment Market: Quantifying Demand

The thesis predicts capital should be flowing from static datasets toward interactive verifiable environments. It is. Per reporting from The Information, Anthropic leadership has discussed spending more than \$1 billion on RL environments over the next year, and a new vendor class (Mechanize, Prime Intellect) has formed around the supply [8]. The unit economics reported by Epoch AI's survey (18 stakeholders) are concrete: task-creation costs run roughly \$200-\$2,000 per task (rising to ~\$20K for complex software/UI replicas), compute spend is ~\$2,400 per

task during RL training, and exclusive deals carry roughly a 4-5x markup over non-exclusive — direct evidence that exclusivity (a proxy for proprietary supply) is independently priced [8].

Incumbent data-labeling firms are pivoting in the same direction. Surge AI reached ~\$1.2B in annualized revenue in 2024 (largely bootstrapped, profitable), working with a concentrated set of frontier labs, and is building RL-environment capabilities [12]. Mercor's valuation reached ~\$10B (Oct 2025, \$350M Series C) with annualized run-rate revenue rising from ~\$75M (Feb 2025) to ~\$450M (around the Series C), explicitly adding RL-environment infrastructure for coding, healthcare, and law [11]. Scale AI took a ~\$14.3B Meta investment for a ~49% stake at a ~\$29B valuation [13]. The broader data-annotation/labeling market is forecast to grow from \$1.2B (2024) to \$10.2B (2034) at a ~23.9% CAGR [10].

Two cautions. First, these are demand-side and supply-side signals, not proof of margin durability in any one vertical. Second, the dominant spend today is in horizontal domains (coding, general agents) where the data is largely public — exactly the lower-defensibility quadrant our thesis flags. The strategic opening is the underexploited high-proprietary, high-verifiable quadrant (regulated clinical, claims/coding, lab/diagnostic workflows).

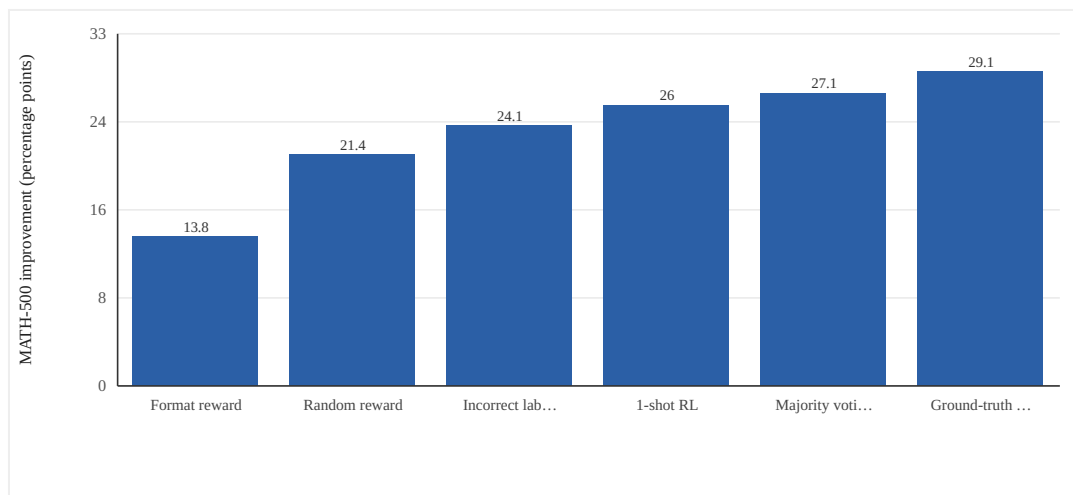


Figure 2. The signal, not the optimizer, carries the RLVR gain (Qwen2.5-Math-7B, MATH-500). On Qwen2.5-Math-7B, RLVR yields large MATH-500 gains even under near-meaningless rewards (random +21.4, incorrect-label +24.1) approaching the ground-truth gain (+29.1). Because the same spurious rewards fail on Llama3/OLMo2, the gain depends on the data/base-model substrate, not the optimizer — direct support that the reward signal is the load-bearing data asset. *Spurious Rewards [3] (arXiv:2506.10947). Improvements over base in absolute points, copied verbatim. · Empirical data.*

6. Analysis

Putting C1-C4 together yields the load-bearing inference: in RLVR, the reward function IS a dataset — a programmatic encoding of which outcomes are correct over a task distribution — and the value of a reasoning model now traces to the quality and uniqueness of that encoding. The spurious-rewards result [3] is the cleanest mechanistic support: vary the optimizer's signal across model families and the gains appear or vanish with the data substrate, not the algorithm. This converts 'environments with checkable outcomes' from a training trick into a manufacturable, defensible data asset.

Why vertical and proprietary specifically? Three forces compound. (1) The data wall: public human text is forecast to be roughly exhausted 2026-2032 [9], so the marginal fresh-signal source is interaction with verifiable environments — and a 2025 result makes the substitution concrete, as jointly training a 1.5B model across the 400

verifiable environments of RLVE-Gym yields +3.37% absolute over six reasoning benchmarks while continuing the original RL run with 3x more compute yields only +0.49%, i.e., scaling environments, not compute, drives the gain [28]. (2) Commoditization of method: GRPO, DAPO, and the R1 distillations are open within months [1,2], the RL-compute-to-performance curve is now itself predictable [27], and even the data-efficiency frontier is moving (a regularized recipe reaches the loss asymptote with 5.17x less data [29]) — so the optimizer is not a moat; the corpus and its checker are. (3) Tail coverage: the open web's rare tails are precisely specialized verticals, where the marginal token (and the marginal verified outcome) is worth most.

The counter-evidence sharpens rather than refutes the thesis. The elicitation-vs-teaching debate — RLVR outperforms the base model at small k but the base model achieves higher pass@k at large k in Yue et al. [18], implying RLVR narrows rather than expands the reasoning boundary — implies RLVR sometimes only surfaces latent base-model skill. If so, building expensive environments for math/code (where skill is plausibly latent and the corpus is public) buys less genuine capability than building them where skill is absent: under-documented proprietary workflows the base models never saw. The debate, followed to its end, is itself an argument for proprietary vertical data.

7. Implications for Data Strategy

Selection rule. Prioritize verticals scoring high on two axes simultaneously: (1) an objective, automatically-checkable outcome (the reward exists), and (2) a proprietary or hard-to-assemble corpus (the data is defensible). Math and code score high on axis 1 but low on axis 2 (public data); regulated clinical claims/coding, insurance adjudication, and lab/diagnostic workflows score high on both [14],[15]. This filter operationalizes the BloombergGPT/AlphaFold template — proprietary data x verifiable outcome — as a forward-looking vertical screen [6,14].

Product shape. The asset is not a static dataset but a re-priced, recurring environment: a corpus plus a checker plus a contamination-free held-out eval that must be refreshed as it leaks. Vendor pricing already reflects this — per-task creation plus per-task RL compute plus an exclusivity premium [8] — favoring a productized environment offering over one-off data dumps.

Build where skill is absent. Given the elicitation risk [3,18], concentrate environment-building where the base model demonstrably lacks the capability (measured on deployment-distribution-matched evals, since aggregate benchmarks mislead). The clinical-coding/claims workflow is the worked high-fit example: a private clinical record paired with a deterministic CDT/ICD/X12 coding/claims outcome — proprietary on axis 2, checkable on axis 1, and outside the base models' training distribution.

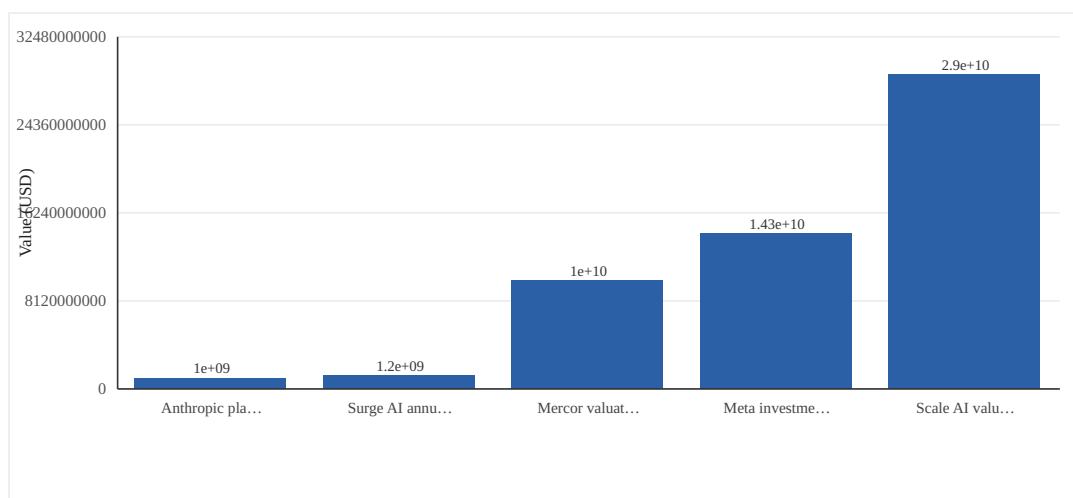


Figure 3. Capital is shifting from static datasets to verifiable environments. Market grounding for the demand side of the thesis. A single lab (Anthropic) has discussed >\$1B on RL environments over a year; environment/data vendors command billion-to-tens-of-billions valuations. These are point-in-time 2025-2026 figures and volatile. *Anthropic spend [8] (The Information via TechCrunch); Surge [12]; Mercor [11]; Scale AI/Meta [13]. Verbatim from reporting. · Empirical data.*

8. Investment Thesis

Building verifiable-reward environments in real, proprietary verticals is simultaneously a training-method bet (RLVR is the frontier recipe — the 2025-2026 open frontier, from DeepSeek-V3.2-Speciale's competition-gold reasoning [26] to predictable RL-compute scaling [27], runs on it), a data-asset bet (the reward is the data, and it survives method churn — and scaling verifiable environments, not compute, is what now moves the needle [28]), and a vertical bet (the defensible supply is proprietary workflows) — three views of one position. The market has validated the demand side at scale: >\$1B planned environment spend over a year by a single lab [8], multi-billion vendor valuations [11,12,13], and a 4-5x exclusivity premium that prices defensibility directly [8].

The differentiated entry is the high-proprietary, high-verifiable quadrant that today's horizontal vendors under-serve. Incumbents (Scale, Surge, Mercor) are racing into coding and general agents — public-data, thin-moat territory. A focused player that secures rights to a proprietary regulated corpus with deterministic outcomes (clinical claims/coding) builds an asset that (a) sidesteps the data wall, (b) cannot be reproduced by scraping, and (c) compounds via a refreshable eval subscription.

Underwriting checklist before committing capital: (1) confirmed legal data-resale/privacy rights to the corpus (the binding gate); (2) a deterministic or expert-adjudicable checker; (3) evidence the base model lacks the skill on a distribution-matched eval; (4) sufficient task supply to avoid single-source fragility. The thesis is strongest where all four hold and weakest where the corpus is public or the rights are unsecured.

9. Limitations and Counter-Evidence

Elicitation vs. teaching. If RLVR primarily elicits latent capability [3,18], the genuine value added by an environment is smaller than headline gains suggest in domains the base model already knows. We mitigate by targeting absent-skill verticals, but this remains an empirical risk that must be measured per target, not assumed.

Benchmark-to-deployment gap (A1). AIME/MATH/MMLU parity [1,2,4] does not guarantee performance on messy clinical or claims workflows; aggregate benchmarks can mislead about where capability is lacking, so

distribution-matched evaluation is a prerequisite, not a nicety. We treat math/code numbers as tractability ceilings, not vertical results.

Reward hacking and signal quality. Over-optimizing a proxy reward follows Goodhart's law — pushing too hard on a proxy reward model harms gold/true-reward performance [25] — so a poorly specified vertical checker can train the wrong behavior. Verifiability is necessary but not sufficient; the checker must be faithful.

Market and rights risk. Cost/valuation figures [8],[10]-[13] are point-in-time and volatile (e.g., market-research forecasts for the annotation market range widely across firms), and the demand multiplier is structural but absolute willingness-to-pay for any specific vertical is unproven. Most decisively, the entire thesis is gated on securing data-resale/privacy rights (e.g., HIPAA-compliant licensing) to the proprietary corpus and on supply scale — without those, a high-fit vertical is unbuildable regardless of its data shape. The scorecard scores the data shape only; the full go/no-go requires the broader investment lens.

10. Conclusion

The reasoning-RL boom rewrote which input is scarce. With pretraining compute commoditizing, the optimizer open-sourced within months, and public text approaching exhaustion [9], the binding constraint on frontier reasoning capability is the verifiable reward signal — and the reward is a dataset. The spurious-rewards result makes the mechanism explicit: capability tracks the data substrate, not the algorithm [3].

The scarce, defensible supply of that signal is proprietary vertical workflows that pair an objective checkable outcome with a hard-to-assemble corpus. The 2025-2026 evidence — open RLVR matching and now surpassing the closed frontier (DeepSeek-V3.2-Special taking IMO and IOI gold with checkable-reward RL) at a fraction of the cost [1,2,26], scaling verifiable environments rather than compute driving the gains [28], RLVR lifting capability in medicine [16,17], and >\$1B of planned environment spend over a year [8] — points one way. Build verifiable-reward environments where the data is proprietary and the base model lacks the skill; that is simultaneously the training, data, and vertical bet, seen from three sides.

INVESTMENT THESIS

In RLVR, the reward function IS the dataset, and the spurious-rewards result is the proof: identical optimization yields +21-24 point MATH-500 gains on Qwen but fails on Llama/OLMo — capability tracks the data substrate, not the algorithm [3]. With open systems now matching and surpassing the closed frontier (DeepSeek-R1 79.8 vs o1 79.2 on AIME 2024, RL phase ~\$294K per the Nature version [1]; DeepSeek-V3.2-Special taking IMO 2025 and IOI 2025 gold and reaching Gemini-3.0-Pro reasoning parity by scaling the same checkable-reward RL [26]), the RL-compute-to-performance curve itself now predictable [27], and a single lab discussing >\$1B on environments over a year [8], the optimizer is commoditized and the scarce input is verifiable signal — and scaling verifiable environments, not compute, is what moves capability (+3.37% from 400 environments vs +0.49% from 3x more compute [28]). The defensible supply is the high-proprietary x high-verifiable quadrant — regulated clinical claims/coding, insurance adjudication, lab/diagnostic — which today's horizontal vendors (chasing public-data coding/agents) under-serve, and where exclusivity already prices at a 4-5x premium [8]. Invest where four gates hold: secured data-resale/privacy rights, a deterministic/adjudicable checker, base-model skill absent on a distribution-matched eval, and sufficient task supply. The clinical-coding bet fits the data shape exactly; the binding risk is rights and supply, not technical feasibility.

References

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- [4] Li, J. et al. (2024). DataComp-LM: In search of the next generation of training sets for language models. arXiv:2406.11794. (DCLM-Baseline 7B reaches 64% 5-shot MMLU with 2.6T tokens; +6.6 pts over MAP-Neo, 40% less compute; model-based filtering 'key to assembling a high-quality training set'.)
- [5] Singhal, K. et al. (2023). Towards Expert-Level Medical Question Answering with Large Language Models (Med-PaLM 2). arXiv:2305.09617. (Up to 86.5% on MedQA; improving upon Med-PaLM by over 19%.)
- [6] Wu, S. et al. (2023). BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564. (50B model; 363B-token proprietary financial corpus + 345B general; outperforms existing models on financial tasks by significant margins.)
- [7] Lambert, N. et al. (2024). Tulu 3: Pushing Frontiers in Open Language Model Post-Training (introduces and names RLVR). arXiv:2411.15124. (RLVR = a novel method training on tasks with verifiable outcomes.)
- [8] Zeff, M. (2025). Silicon Valley bets big on 'environments' to train AI agents. TechCrunch, Sep 21, 2025 (citing The Information for Anthropic >\$1B/yr; Mechanize, Prime Intellect); and Epoch AI, An FAQ on Reinforcement Learning Environments (epoch.ai/gradient-updates/state-of-rl-envs) (~\$2,400 compute/task; \$200-\$2,000 task creation; 4-5x exclusivity premium; based on 18 stakeholder interviews).
- [9] Villalobos, P. et al. (2022/2024). Will we run out of data? Limits of LLM scaling based on human-generated data. arXiv:2211.04325 / ICML 2024. (Models trained on datasets ~equal to the stock of public human text between 2026 and 2032.)
- [10] Global Insight Services, Data Annotation and Labeling Market (2025). Forecast \$1.2B (2024) -> \$10.2B (2034), ~23.9% CAGR. Third-party projection, not realized revenue; other market-research firms report materially different figures.
- [11] Coverage of Mercor's RL-environment pivot and financials (2025): ~\$10B valuation (Oct 2025, \$350M Series C led by Felicis); annualized run-rate ~\$75M (Feb 2025) -> ~\$450M (Sep/Oct 2025); adding RL-environment infrastructure for coding, healthcare, law. (TechCrunch, Sep 9 & Oct 27, 2025.)
- [12] Surge AI financials and RL-environments organization (2024-2025): ~\$1.2B annualized revenue in 2024, largely bootstrapped and profitable, concentrated frontier-lab customers (OpenAI, Google, Anthropic, Meta); building RLHF/RL-environments capability. (Inc., Lenny's Newsletter, Sacra, 2025.)
- [13] Scale AI / Meta deal (2025): Meta ~\$14.3B investment for ~49% stake; Scale AI valuation ~\$29B (up from ~\$13.8B, May 2024); Alexandr Wang to Meta. (TechCrunch, Jun 13, 2025.)
- [14] Jumper, J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature 596. (Trained on experimentally-resolved structures; median 92.4 GDT and above-90 GDT for ~two-thirds of targets at CASP14.)
- [15] Chen, Z. et al. (2023). MEDITRON-70B: Scaling Medical Pretraining for Large Language Models. arXiv:2311.16079. (Within 5% of GPT-4 and 10% of Med-PaLM-2 on major medical benchmarks via curated corpus; continued pretraining on Llama-2.)
- [16] Med-RLVR: Emerging Medical Reasoning from a 3B base model via Reinforcement Learning (2025). arXiv:2502.19655. (3B base; matches SFT in-distribution; +8-point OOD accuracy gain.)
- [17] Rubrics as Rewards: Reinforcement Learning Beyond Verifiable Domains (2025). arXiv:2507.17746. (Best RaR variant: up to +31% relative improvement on HealthBench vs Likert baseline; +7% on GPQA-Diamond.)
- [18] Yue, Y. et al. (2025). Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837. (RLVR outperforms base at small k but base achieves higher pass@k at large k — RL narrows rather than expands the boundary; reasoning bounded by the base model.)
- [21] Ouyang, L. et al. (2022). Training language models to follow instructions with human feedback (InstructGPT). arXiv:2203.02155. (Canonical SFT -> reward model -> PPO RLHF pipeline.)
- [22] Cobbe, K. et al. (2021). Training Verifiers to Solve Math Word Problems (GSM8K). arXiv:2110.14168. (Verification scales more effectively with added data than a finetuning baseline.)
- [23] Lightman, H. et al. (2023). Let's Verify Step by Step. arXiv:2305.20050. (Process supervision significantly outperforms outcome supervision on MATH; PRM solves 78% of a representative subset of the MATH test set; releases PRM800K.)

- [24] Hoffmann, J. et al. (2022). Training Compute-Optimal Large Language Models (Chinchilla). arXiv:2203.15556. (Current large models significantly undertrained; model size and tokens should scale equally.)
- [25] Gao, L., Schulman, J., Hilton, J. (2022). Scaling Laws for Reward Model Overoptimization. arXiv:2210.10760. (Over-optimizing a proxy reward model harms gold-reward performance, in accordance with Goodhart's law.)
- [26] DeepSeek-AI (2025). DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models. arXiv:2512.02556. (Scaling post-training RL compute brings V3.2 to GPT-5 parity; the high-compute V3.2-Special variant surpasses GPT-5 and matches Gemini-3.0-Pro, winning gold at IMO 2025 (35/42) and IOI 2025 (492/600); DeepSeek Sparse Attention makes the scaled RL post-training tractable.)
- [27] Khatri, D., Madaan, L. et al. (2025). The Art of Scaling Reinforcement Learning Compute for LLMs (ScaleRL). arXiv:2510.13786. (First >400,000-GPU-hour study fitting sigmoidal compute-performance curves for RL; most design choices modulate compute efficiency without shifting the asymptote, while the recipe does shift it — validated on a single 100,000-GPU-hour run.)
- [28] Zeng, Z., Ivison, H., Wang, Y., Yuan, L. et al. (2025). RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments. arXiv:2511.07317. (RLVE-Gym = 400 verifiable environments that procedurally generate problems and adapt difficulty; joint training across all 400 yields +3.37% absolute over six reasoning benchmarks on a 1.5B model, versus +0.49% from 3x more compute on the original data — environment scaling, not compute, drives the gain.)
- [29] Kim, K., Kotha, S., Liang, P., Hashimoto, T. (2025). Pre-training under infinite compute. arXiv:2509.14786. (A regularized + ensembled recipe reaches the validation-loss asymptote with 5.17x less data than baseline, with optimal weight decay ~30x larger than standard — the data-efficiency frontier is moving fast as the public-text stock approaches exhaustion.)